Cambrian College

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality Course Outline

Cambrian College would like to acknowledge that we are situated on the traditional lands of the Robinson Huron Treaty, and acknowledge our host, Atikameksheng Anishnawbek, as well as our ancestors of this land, and the Anishnaabe people.

Course Title	Stats and Data Visualization				
Course Code:	QMM1002	Credit Value:	4	Credit Hours:	56
Programs:	BAPG Business Analytics CAGC Crime Analytics CAPT Crime Analytics - Pt HAGC Health Analytics HAPT Health Analytics - Pt				
Equivalencies		Prerequisites		Corequisites	

This course may be delivered in a variety of different formats: 100% in-class, 100% online (or a blend of both), videoconferencing, distributed learning or off-campus. Please confirm with your faculty member which format will be used for your section of this course.

General Education Course: ☐ Degree Breadth Course: ☐ Eligible for PLAR: ☑

COURSE DESCRIPTION

In this course, students will build on the fundamental statistical tools, such as confidence intervals, hypothesis testing and time series analysis,
and further explore how these tools are used to process and analyze data. Using the statistical package R, students will focus on how to use and interpret data output and how to visualize results.

Date: May 25, 2021

Prepared By: Sidney Shapiro

Approved by:

JILL FERGUSON
Chair, School Of Business And Information Technology, Creative Arts, Design, Music And Hospitality

Effective: Fall 2021, Winter 2022, Spring 2022

RELATIONSHIP TO PROGRAM VOCATIONAL LEARNING OUTCOMES

PROGRAM LEVEL	
This course contributes to your program by allowing you to demonstrate the following vocational learning outcomes:	
Program(s)	**Vocational Learning Outcomes**
Business Analytics	**College Standards** 1. Design and apply data models that meet the needs of a specific operational/ business process. 2. Design and present data visualizations to communicate information to business stakeholders. 3. Apply data analytics, business intelligence tools and research to support evidence-based decision making. 4. Identify and assess data analytics business strategies and workflows to respond to new opportunities or provide project solutions.
Crime Analytics	**College Standards** 1. Design and apply data models that meet the needs of a specific operational process. 2. Design and present data visualizations to communicate information to stakeholders. 3. Create effective oral presentations and produce fitting analytical products suitable for specific audiences.
Crime Analytics - Pt	**College Standards** 1. Design and apply data models that meet the needs of a specific operational process. 2. Design and present data visualizations to communicate information to stakeholders. 3. Create effective oral presentations and produce fitting analytical products suitable for specific audiences.
Health Analytics	**College Standards** 1. Collect and analyze a variety of data sets to identify information related to organizational health needs. 2. Design health indicators and apply data models that meet organizational, program and service needs. 3. Mine and manipulate data sets, correlate information and produce outputs using advanced software functions. 4. Design visualizations and communicate results of data analysis to decision makers, stakeholders and the public. 5. Apply health analytics and business intelligence tools to support evidence-informed practice.
Health Analytics - Pt	**College Standards** 1. Collect and analyze a variety of data sets to identify information related to organizational health needs.

Program(s)	Vocational Learning Outcomes
	2. Design health indicators and apply data models that meet organizational, program and service needs. 3. Mine and manipulate data sets, correlate information and produce outputs using advanced software functions. 4. Design visualizations and communicate results of data analysis to decision makers, stakeholders and the public. 5. Apply health analytics and business intelligence tools to support evidence-informed practice.

COURSE CURRICULUM

Topics/Concepts Covered in This Course

- Confidence intervals and hypothesis tests for means
- Comparing two means
- Design of experiments
- Analysis of variance
- Chi-square tests
- Time series analysis

COURSE LEVEL: Learning Outcomes and Objectives	
To earn credit for this course, you must reliably demonstrate your ability to:	
Learning Outcome	**Objectives**
1. Construct, utilize, and interpret confidence intervals for means.	1.1 Perform calculations involving the sampling distribution of the mean. 1.2 Construct and interpret confidence intervals for means. 1.3 Identify the relationship between sample size and a confidence interval. 1.4 Compute sample sizes for confidence intervals for means. 1.5 State and check the assumptions for confidence intervals. 1.6 Investigate normality by creating boxplots and histograms of data using R.
2. Perform and interpret a hypothesis test for a sample mean.	2.1 Formulate and explain the null and alternative hypotheses 2.2 Perform a hypothesis test for a mean using the p-value method and critical-value method and interpret the results. 2.3 Describe and interpret p-values and alpha levels. 2.4 Identify when to perform a one-sided or two-sided hypothesis test.

Learning Outcome	Objectives
	2.5 Classify Type I and Type II errors. 2.6 State and check the assumptions for hypothesis testing for a sample mean. 2.7 Calculate p-values and critical t-values with R.
3. Use the two-sample t-test to compare two sample means.	3.1 Carry out calculations involving the sampling distribution for the difference between two means. 3.2 Formulate the null and alternative hypotheses. 3.3 Perform a two-sample t-test using the p-value method and critical value method and interpret the results. 3.4 State and check the assumptions for a two-sample t-test. 3.5 Determine a confidence interval for the difference between two means. 3.6 Calculate the appropriate number of degrees of freedom for a two-sample t-test. 3.7 Create bar-plots of group means with error-bars using R.
4. Carry out and interpret the results of pooled and paired t-tests.	4.1 Determine whether data are paired or unpaired. 4.2 Identify when it is appropriate to use pooled and paired t-tests. 4.3 State and check the assumptions for pooled and paired t-tests. 4.4 Indicate conditions under which the homogeneity of variance assumption may be satisfied. 4.5 Perform both pooled and paired t-tests using the p-value method and critical value method, and interpret the results. 4.6 Calculate pooled and paired t-confidence intervals. 4.7 Create a grouped pairwise-differences bar-plot using R for paired data.
5. Explain the core concepts of experimental design and analysis of variance (ANOVA).	5.1 Compare and contrast observational studies with experiments. 5.2 Given a design, identify the factor(s), level(s), treatment(s), and response(s). 5.3 State and describe the four principles of experimental design. 5.4 Differentiate between completely randomized, randomized block, and factorial designs. 5.5 Explain the use of blinding and placebos in experiments,and determine when they are appropriate. 5.6 Identify when a confounding or lurking variable may be present. 5.7 Calculate the family-wise error rate, and use it to explain why ANOVA is beneficial.

Learning Outcome	Objectives
	5.8 Determine when to use the ANOVA, and describe the assumptions, conditions, and F-ratio. 5.9 State the null and alternative hypotheses for the ANOVA.
6. Compare two or more groups using a one-way ANOVA and interpret the results.	6.1 Perform a one-way ANOVA and interpret the results 6.2 Explain the meaning of a statistically significant F-ratio 6.3 Utilize the Tukey HSD test to evaluate pairwise comparisons of means, and explain the results. 6.4 Compute p-values and critical F-values using R. 6.5 Create boxplots for group data and bar-plots with error-bars to summarize group mean differences.
7. Perform a two-way ANOVA to explain the effect of two factors on a response variable.	7.1 Given an interaction plot, identify any potential main effects and interactions. 7.2 State the null and alternative hypotheses for the two-way ANOVA. 7.3 Carry out a two-way ANOVA, summarize the results in an ANOVA table, and interpret the F-ratios. 7.4 Utilize the Tukey HSD test to identify which mean comparisons are statistically significant. 7.5 Compute p-values and critical F-values using R. 7.6 Create and interpret an interaction plot. 7.7 Use boxplots and bar-plots to summarize data and group means.
8. Utilize the chi-square goodness of fit test to measure the fit of a sample of data to an expected distribution.	8.1 Identify when it is appropriate to use a chi-square goodness of fit test. 8.2 Define the null and alternative hypotheses for the chi-square goodness of fit test. 8.3 Calculate the expected frequencies and chi-square statistic, and then test for its significance and interpret the result. 8.4 Use residuals to compare the observed and expected distributions. 8.5 State and verify the assumptions and conditions for the chi-square goodness of fit test. 8.6 Create a grouped bar-plot of the observed and expected distributions
9. Perform a chi-square test for independence and interpret the results.	9.1 Identify whether or not the chi-square test for independence is appropriate for a given design. 9.2 State the null and alternative hypotheses for the chi-square test for independence. 9.3 Calculate the expected frequencies and chi-square statistic, and then test for its significance and interpret the results.

Learning Outcome	Objectives
	9.4 Compute the standardized residuals and use them to identify differences between the observed and expected distributions 9.5 Create and interpret association plots and mosaic plots for bivariate categorical data 9.6 Create a grouped stacked bar-plot to visualize differences between observed and expected frequencies across groups.
10. Describe the basic properties of time series and use the moving average method to produce forecasts.	10.1 Define a time series. 10.2 Calculate index numbers and use them to compare time series. 10.3 Explain the trend, seasonal, cyclic, and irregular components of a time series. 10.4 Decompose a time series into its constituent components 10.5 Fit a moving average model of specified length to a sample of time series data. 10.6 Use a moving average model to forecast a time series. 10.7 Describe how the length of the moving average affects the properties of the model. 10.8 Compute and interpret forecast errors (FE), mean squared error (MSE), mean absolute deviation (MAD), and mean absolute percentage error (MAPE). 10.9 Create plots of time series and moving average models.
11. Use simple, Holt, and Holt-Winters exponential smoothing models to forecast a time series.	11.1 Compute and interpret the parameters for simple, Holt, and Holt-Winters exponential smoothing models. 11.2 Fit a simple, Holt, and Holt-Winters exponential smoothing model to a given time series. 11.3 Forecast a time series using simple exponential smoothing, Holt exponential smoothing, and Holt-Winters exponential smoothing 11.4 Summarize the forecast accuracy using the FE, MSE, MAD, and MAPE 11.5 Determine which forecasting method is appropriate for a given time series. 11.6 Create plots of time series, exponential smoothing models, and confidence bounds for forecasted values.

Essential Employability Skills

Communication

- communicate clearly in written, spoken, and visual form that fulfills purpose/needs of audience.

Numeracy

- execute mathematical operations accurately.

Critical Thinking and Problem Solving

- apply a systematic approach to solve problems.

Information Management
- locate, select, organize, and document information using appropriate technology and info systems.
- analyze, evaluate, and apply relevant information from a variety of sources.

Interpersonal
- show respect for the diverse opinions, values, belief systems, and contributions of others.

Personal
- take responsibility for one's own actions, decisions, and consequences.

Delivery Method

- Classroom: Course is delivered through scheduled synchronous teaching that may be face-to-face and/or virtual.
- Online: Course is fully delivered through asynchronous teaching.
- Hybrid: Course combines scheduled synchronous and unscheduled asynchronous teaching.
- HyFlex: Course includes both synchronous and asynchronous learning and the student can move between both components seamlessly.

Learning Activities

- Lectures
- Class Discussions
- Labs
- In-Class Exercises
- Case Studies
- eLearning Components

Resources Required

Books

Norean R. Sharpe Richard D. De Veaux Paul F. Velleman David Wright David Wright, *Business Statistics, Fourth Canadian Edition, 4th edition*, 4th, Pearson
ISBN: 9780136964032

Additional Supplies

TEXTBOOK
Students may choose to purchase an older edition of the textbook listed above, please contact the course instructor for more details

Evaluation Plan

Grading Scheme

A	80% - 100%
B	70% - 79%
C	60% - 69%
D	50% - 59%
F	0% - 49%

Evaluation Method	Value (%)
Assignments	**40%**
Please see your instructor for details as the assignments may be in the form of written case studies, presentations, individual or group projects, or another form not listed.?	

Evaluation Method	Value (%)
Applied Activities (G)	**60%**
There will be assignments covering the material for module which will incorporate R software as determined by the instructor.	

ADDITIONAL INFORMATION

College

Modifications to the Course Evaluation Plan
Under exceptional circumstances, Cambrian reserves the right to alter the Course Evaluation Plan. Upon approval from the Dean, faculty members will notify students of any modification to the Course Evaluation Plan and post to the course Learning Management System site (Moodle).

Academic Policies
Students must adhere to all academic policies. These are available on myCambrian as well as the College website at https://cambriancollege.ca/about/official-documents-and-policies/academic-policies/.

Accessibility Services
Cambrian is committed to creating an inclusive learning environment where we support our students" success. Students who are registered with the Glen Crombie Centre can access accessibility and/or counselling services by:

1. Registering and scheduling an appointment through Clockwork in the Student Portal in myCambrian.
2. Contacting 705-566-8101 ext. 7420 or 7311, or
3. E-mailing accessibilityservices@cambriancollege.ca or Counselling@cambriancollege.ca

Attendance
Regular attendance is strongly encouraged as it contributes to student success. Some courses have specific lab/participation requirements. Students are expected to be aware of, and adhere, to these requirements.

Audio/Visual Capture
Sounds and images from this class, and contributions made by a participant, virtually or in-person, are recorded under the authority of the Ontario Colleges of Applied Arts and Technology Act, 2002. The main purpose for these recordings is to allow students enrolled in this course, whether they attend the particular class in person, virtually, or at all, to review class content and activity. The use of class recordings is for personal use only and shall not be shared or transferred.

These recordings may also be reviewed by faculty members to prepare future classes, to evaluate students, to collaborate in program/course development/review, or to provide feedback to students or other faculty. Any questions about this collection may be addressed to your respective Dean.

Copyright
Students may not submit work or partake in activities that infringe on the Canadian Copyright Act, or the related Cambrian College Fair Dealing Guidelines and Copying Guidelines. The guidelines may be viewed at: http://cambriancollege.libguides.com/copyrightinfo or by contacting the Library for questions related to copyright: library@cambriancollege.ca

Wabnode Centre for Indigenous Services
Students are encouraged to come to Wabnode to meet our Elders and to access personal counselling and connect with academic tutoring. Our team is also be able to provide students with information related to funding and scholarship and bursary opportunities for Indigenous students. We encourage students to come to Wabnode and find out about opportunities to share in cultural, sports and community activities. Students can also contact Wabnode at wabnode@cambriancollege.ca

Out-of-Class Assistance
The faculty will inform students of their Out-of-Class availability through the Learning Management System site (Moodle).

Prior Learning Assessment and Recognition (PLAR)
Students wishing to have work or life experience considered through Prior Learning Assessment and Recognition should contact pathways@cambriancollege.ca

Testing Practices at Cambrian
Many courses include major tests and/or final exams. The practice at Cambrian requires that these types of test situations involve proctoring to ensure academic integrity. Online tests/exams may employ a proctoring service to enable you to take your exam from a location of your choosing within a period specified by your instructor. When you are taking an online test/exam, the proctoring service may capture your video, screen, audio and web surfing particulars to protect academic integrity. Cambrian College collects, uses, discloses and retains personal information in compliance with the Freedom of Information and Protection of Privacy Act (FIPPA). Your personal information is being collected under the authority of the Ontario Colleges of Applied Arts and Technology Act S.O. 2002, c.8, Sched. F. This information will be used for the purpose of administering a test/exam through an online proctoring service acting as an authorized agent of the College. Please refer to Cambrian's Confidentiality of Student Records Policy for more details. If you have any questions regarding the collection of your personal information, please contact Vice President Academic, Cambrian College, 1400 Barrydowne Rd., Sudbury ON P3A 3V8, 1-705-566-8101.

Tests and Evaluations
Students are required to write tests and/or complete evaluations as scheduled. Exceptions may be made in the event of an emergency or a sanctioned event (e.g., varsity sports, field trip, religious observances, etc.). In non-emergency situations, students are expected to contact the faculty member in the event that they can not be present for a test/evaluation to explain the reason for their absence. At the discretion of the faculty member, an alternate time for the test/evaluation may be allowed.

Transfer Credit
Students wishing to have courses from other programs or institutions assessed for equivalency and/or transfer credit should contact transfercredits@cambriancollege.ca.

Equity, Diversity and Inclusivity
Cambrian is committed to building and preserving an equitable, diverse, and inclusive learning community where students, faculty, and staff may achieve their full potential in an environment characterized by equality of respect and opportunity. All students and employees have the right to live and work in an environment that is free from discrimination and harassment. Therefore, Cambrian College will not tolerate any form of discrimination or harassment in its employment, education, accommodation, or business dealings. For more information, please visit: https://cambriancollege.ca/about/official-documents-and-policies/equity-human-rights-and-accessibility/